



← Projects



Personal Portfolio using Astro



Nov 2025 – Jan 2026



Associated with Genworth

This project contains the source code for my personal portfolio website, built to present my professional experience, engineering projects, technical skills, and certifications in a clean, fast, and modern web experience.

The site serves as my primary professional landing page for:

- Recruiters and hiring managers
- Engineering collaborators
- Technical reviewers

It focuses on:

- Clear storytelling of technical experience
- Clean, readable UI/UX
- High performance and accessibility
- Maintainable, component-driven architecture

Skills: Astro, JavaScript, +4 skills



Alan Palayil — Personal Portfolio

The portfolio showcases:

- Professional experience and roles...

Personal Portfolio using React Native



Jan 2023 – Nov 2025



Associated with Illinois Institute of Technology

This project is a production-grade personal portfolio website built to present my professional experience, technical projects, and certifications in a clear, structured, and visually polished format.

The site is implemented using React (React Native Web), JavaScript, HTML, and CSS, and is designed with a strong emphasis on:

- Clean, readable UI/UX
- Responsive layout across devices
- Maintainable component-based architecture
- Performance and usability
- Clear storytelling of technical experience and projects

Skills: Git, JavaScript, +3 skills

Enhancing Plant Health with AI



May 2023 – Aug 2023



Associated with Illinois Institute of Technology

I was part of a project focused on addressing the critical challenge of plant diseases and their impact on global food security. Our innovative solution integrates deep learning techniques with environmental sensors to create a comprehensive system for plant disease detection and classification. By combining thermal imaging technology and humidity sensors, we capture vital environmental data, enabling accurate disease assessment. Our convolutional neural network (CNN) model, trained on a diverse dataset of plant images, effectively identifies and classifies various diseases, offering real-time detection capabilities. This groundbreaking approach not only aids in early risk identification but also provides insights into irrigation needs through Vapor Pressure Deficit (VPD) calculations. Our solution holds promise for sustainable agriculture, aiding farmers, agronomists, and researchers in combatting plant diseases and ensuring resilient crop production.

Skills: Artificial Intelligence (AI), Python (Programming Language), +3 skills



Enhancing Plant Health with AI Repository

Keywords: Plant disease detection, deep learning, convolutional neural networks, environmental sensors, thermal imaging, Vapor Pressure Deficit (VPD), disease ...

Other contributors



32-bit RISC Processor: Design and Implementation with MIPS Pipeline and Hazard Detection



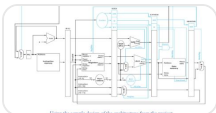
Aug 2022 – Dec 2022



Associated with Illinois Institute of Technology

The project involved creating a custom 32-bit RISC processor based on the MIPS processor. The design included implementing a 5-stage pipeline, including instruction fetch, decode, execute, memory access, and write-back. The team referred to a minimum subset of ISA to create their own clock cycle per stage, equally assigning them between each stage. They also implemented a forward unit and hazard detection unit to overcome hazards in test bench data inputs. The project required a combination of technical skills, including processor design, pipeline implementation, and hazard detection.

Skills: RISC-V, MIPS Assembly, +1 skill



32-bit RISC Processor: Design and Implementation Repository

Keywords: MIPS Architecture, VHDL, Verilog, RISC-V, Pipelining, ISA Standards

Other contributors



Convo Care: Mental Health Sentiment Analysis Tool for Healthcare Workers



Aug 2022 – Dec 2022



Associated with Illinois Institute of Technology

The project aimed to develop a web-based sentiment analysis model to help healthcare professionals monitor and assess the mental health of students. The project utilized various technologies such as Python, JavaScript, CSS, React Database, VOIP, and MySQL. Python was used to develop the sentiment analysis model that analyzed text and voice conversations between students and healthcare professionals. JavaScript and CSS were used to create the web interface for healthcare professionals to view the mental health scores. React Database was used to store student data and mental health scores, and MySQL was used as the database management system.

Skills: Python (Programming Language), PostgreSQL, +2 skills

Other contributors



SensoryPi: A Comprehensive Smart Security Alarm System with Facial Recognition and Motion Detection



May 2022 – Jul 2022



Associated with Illinois Institute of Technology

Show project

The project aimed to create a smart security alarm system with a large coverage area, power backup, motion detection sensors, and facial recognition front interface. The system utilized Raspberry Pi and other sensory nodes and was designed to provide an added layer of security to its users. The team utilized several technologies, including Ubuntu, Python libraries, Arduino, OpenCV, and MQTT platform, to achieve the project's objectives. The project required a combination of technical skills, including hardware design, programming, and image processing. The successful completion of the project resulted in the implementation of an innovative and comprehensive smart security alarm system that made it an exciting addition to the field of security systems.

Skills: Mqtt, Project Management, +4 skills

Other contributors



Through The Speculum

Jan 2022 – May 2022

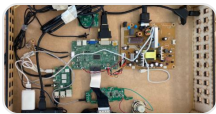


Associated with Illinois Institute of Technology

Show project

The project created an interactive mirror that could be customized by users to control multiple services and integrate with smart homes. The mirror was powered by a Raspberry Pi and designed to support touch and gesture control. To implement the project, the team utilized Python libraries, Arduino, OpenCV, and Android development. The project required a combination of technical skills, including programming, hardware design, and user interface design. The successful completion of the project provided an innovative and user-friendly solution for individuals looking to control multiple services and integrate with their smart homes.

Skills: Mqtt, Project Management, +4 skills



Smart Mirror - Through the Speculum Demonstration - ECASP @ Illinois Tech

The Speculum Smart Mirror is not only a tool that will be used on a daily basis for checking one's appearance. Rather than providing that limited functionality, our...

Other contributors



Virtual Golf Club Fitting Solution for Wilson Sports

Aug 2021 – Dec 2021

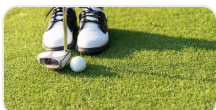


Associated with Illinois Institute of Technology

Show project

The project aims to provide a virtual and remote approach for the process of undergoing a custom golf club fitting. This is achieved through a website and demo kits with attached sensors, which help to narrow down the best-suited clubs for a beginner or intermediate golfer. The project offers an innovative solution to the challenge of custom golf club fitting, making it more accessible and convenient for golfers to improve their game.

Skills: Collaborative Problem Solving, Web Development



Wilson Custom Golf Solution

Website design for final presentation!

Other contributors





Hawk Study using Java Maven



Jan 2021 – May 2021



Associated with Illinois Institute of Technology

Show project

The IPRO 497 course involved an advanced practical project aimed at developing a study room reservation system that adheres to Covid-19 guidelines for students to study on campus. The project utilized several functions, including Java Maven, SpringBoot, PostgreSQL, and a ThingSpeak server to ensure efficient and effective system management.

Skills: Java, Git, +5 skills

Other contributors



Office Supply Order Management System using Java



Aug 2019 – Dec 2019



Associated with Illinois Institute of Technology

Show project

This project implements a small order intake and processing engine in Java that simulates how enterprise systems ingest, validate, and manage structured transaction data from external sources.

The system reads order records from an input file, validates them against a set of business and data-quality rules, converts them into strongly-typed domain objects, and provides basic operations for querying and managing the in-memory dataset.

The project models two types of transactions:

One-time orders

Recurring orders

Both share a common base type and are processed using polymorphic logic, similar to how real-world financial and operations systems handle different transaction classes through a unified pipeline.

Skills: Java, Data Validation, +1 skill

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